(v) Favourable Events:

The number of cases favourable to an event in a trial is the number of outcomes which entail the happening of the event.

(vi) Mutually Exclusive events:

Events are said to be mutually exclusive if the happening of any one of them precludes the happening of all the others, ie., if no two or more of them can happen simultaneously in the same trial. Symbolically the event A and B are mutually exclusive if $A \cap B = \varnothing$.

(vii) Equally Likely Events:

Events (two or more) of an experiment are said to be equally likely, if any one of them cannot be expected to occur in preference to the others.

(viii) Classical definition of Probability

If a random experiment or trial results in 'n' exhaustive, mutually exclusive and equally likely outcomes (or cases), out of which m are favourable to the occurrence of an event E, then the probability 'p' of occurrence (or happening) of E, usually denoted by $P(E)$, is given by

$$P = P(E) = \frac{Number\ of\ favourable\ cases}{Total\ number\ of\ exhastive\ cases} = \frac{m}{n}$$



James Bernoulli who was the first person to obtain a quantitative measure of uncertainty.

(ix) Properties

(i) $0 \le P(E) \le 1$

(ii) Sum of all the probability equal to 1.

(iii) If $P(E) = 0$ then E is an impossible event.

For example : A coin is tossed. Find the probability of getting a head.

Solution: The total possible outcomes of an experiment {H,T}

Therefore $n = 2$

The favourable outcome for getting a head {H}. Therefore m =1. Thus the required probability is

$$P(\text{getting a } H\} = \frac{m}{n} = \frac{1}{2}$$

(x) Modern Definition of Probability

The modern approach to probability is purely axiomatic and it is based on the set theory concepts. In order to study, the theory of probability with an axiomatic approach it is necessary to define certain basic concepts. They are

(i) **Sample Space:** Each possible outcome of an experiment that can be repeated under similar or identical conditions is called a sample point and the collection of sample points is called the sample space, denoted by S.

(ii) **Event:** Any subset of a sample space is called an event.

(iii) **Mutually Exclusive events:** Two events A and B are said to be mutually exclusive events if $A \cap B = \phi$ i.e., if A and B are disjoint sets.

Example: Consider S = { 1,2,3,4,5}

Let A = the set of odd numbers = {1,3,5}

and B = the set of even numbers = {2,4}

Then $A \cap B = \phi$

(iv) Therefore the events A and B are mutually exclusive.

(xi) Observation:

Statement meaning in terms of Set theory approach.

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